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Use Cases:

1. Title: Take input data

Main Success Scenario:

- a. User selects option to manually input a dataset
- b. User is then given two columns and inputs the desired data into the columns
- c. System validates the dataset
- d. User then proceeds to chart generation

Extensions:

B1: User does not input any data

- A. System validates the dataset and finds no data
- B. System kicks user back to input and prompts them to enter data to be converted to a chart

C1: System finds invalid data (such as 3/0 or other invalid types)

- A. System reports the error with the invalid point and a message (cannot divide by 0, irrational number, etc)
- B. User is prompted to fix the error

2. Title: Translate input data based off of input type (JSON/YAML etc to native type)

Main Success Scenario:

- a. User opens the system and selects any option to create a chart
- B. System will ask the user to supply a data file
- C. User supplies a data file organized in two columns of data points
- D. System recognizes the file type and translates the data accordingly for use in the chart
- E. User then progresses to the chart options with the given dataset

Extensions:

D1. User supplies an unsupported filetype (exe, other)

- i. System recognizes it as a non data file
- ii. System sends a warning message to the user asking for a valid datatype
- iii. User must start back from beginning of sequence

D2. Recognizing the file type

- i. System looks at the extension name
- ii. System matches the extension to a translation type
- iii. System then translates the data with the respective translation system

3. Title: Generate Plot (line chart, scatterplot, or other)

Main Success Scenario:

- a. User selects option to create scatterplot
- b. System asks user what variables should be used for x and y axis
- c. System maps selected variables to proper axis
- d. System creates graph/plot

- e. Scatterplot is displayed for user

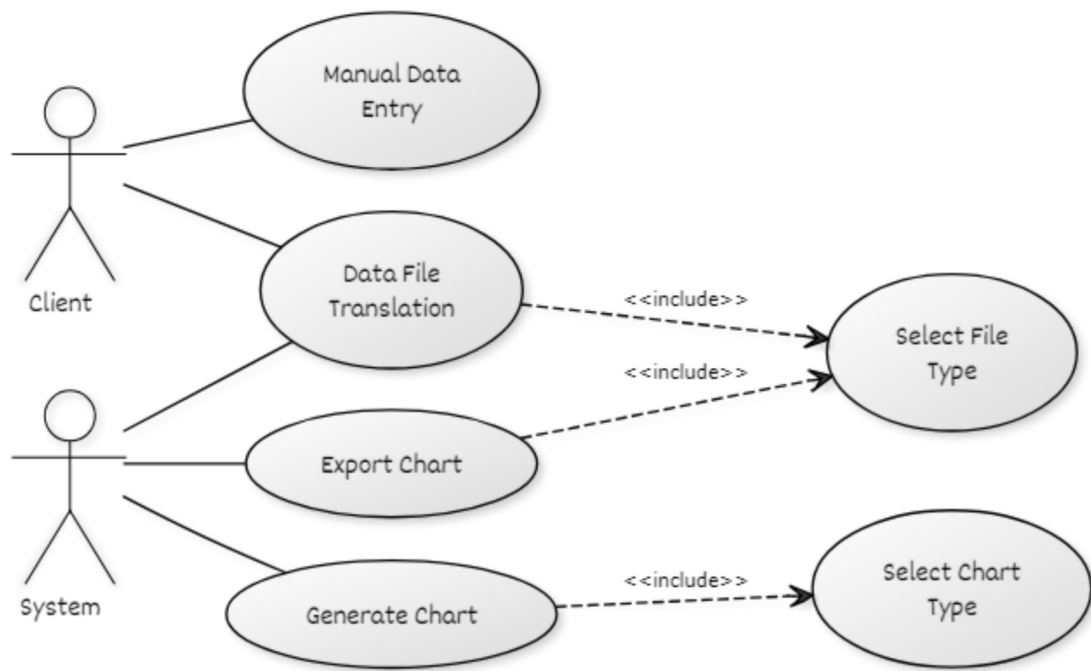
Extension:

- 2a. User selects bar graph
- 2b. System asks user what variables should be used for x and y axis
- 2c. System checks if x axis is a continuous variable
- 3c. If continuous, ask user if sure
- Return to step d
- 4. Title: Output chart image (PDF/SVG)
 - a. User is prompted to select whether they would like to export this chart to an image file
 - b. User is prompted to choose a type of image type to save the chart as
 - c. System takes the selected image type and converts the chart to the respective type (pdf, svc, etc.)
 - d. User then receives the converted file, and is prompted to save it

Extensions

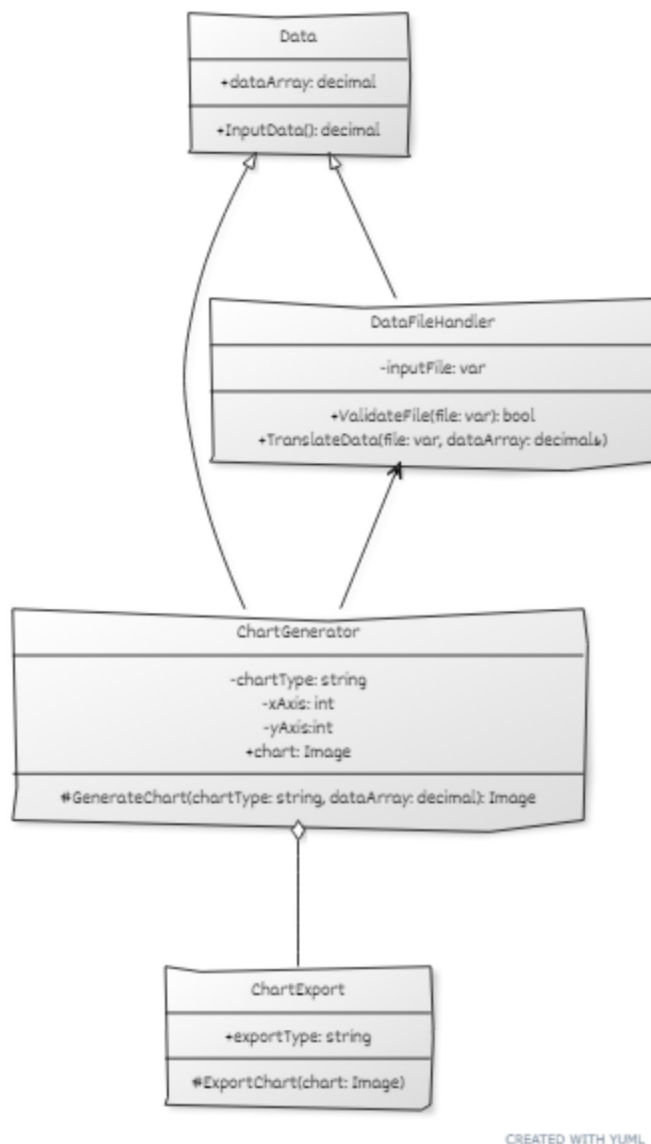
- B2: Selected type is not supported
 - a. The user is kicked back to the file selection screen and given an error message
 - b. The user then selects a different file type
 - c. Repeat until a valid choice is selected
- D2: Automatic download setting
 - a. If an option to automatically download generated charts is selected, the system does not prompt the user and instead automatically downloads the file to the user's computer.

Use Case Diagram:



CREATED WITH YUML

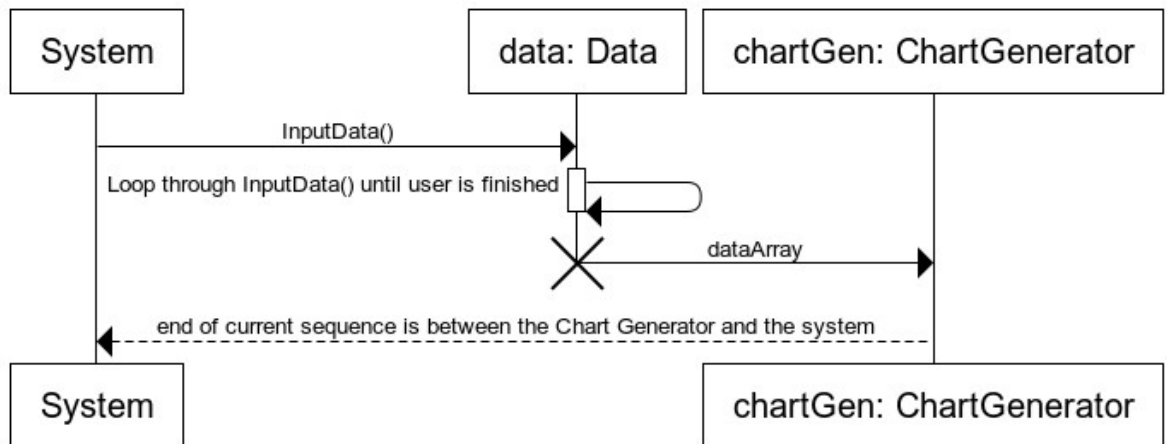
Class Diagram:



Sequence Diagram:

Use Case 1: User Inputs Data Manually

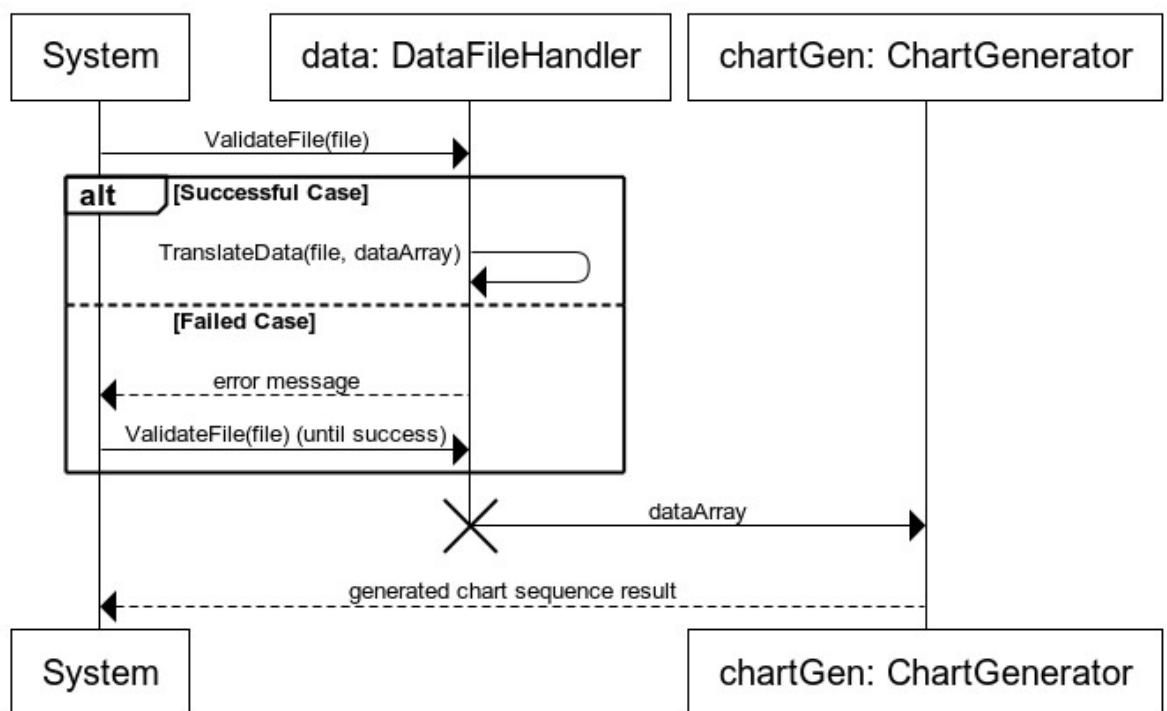
Manual User Data Input



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Use Case 2: Translating Data File

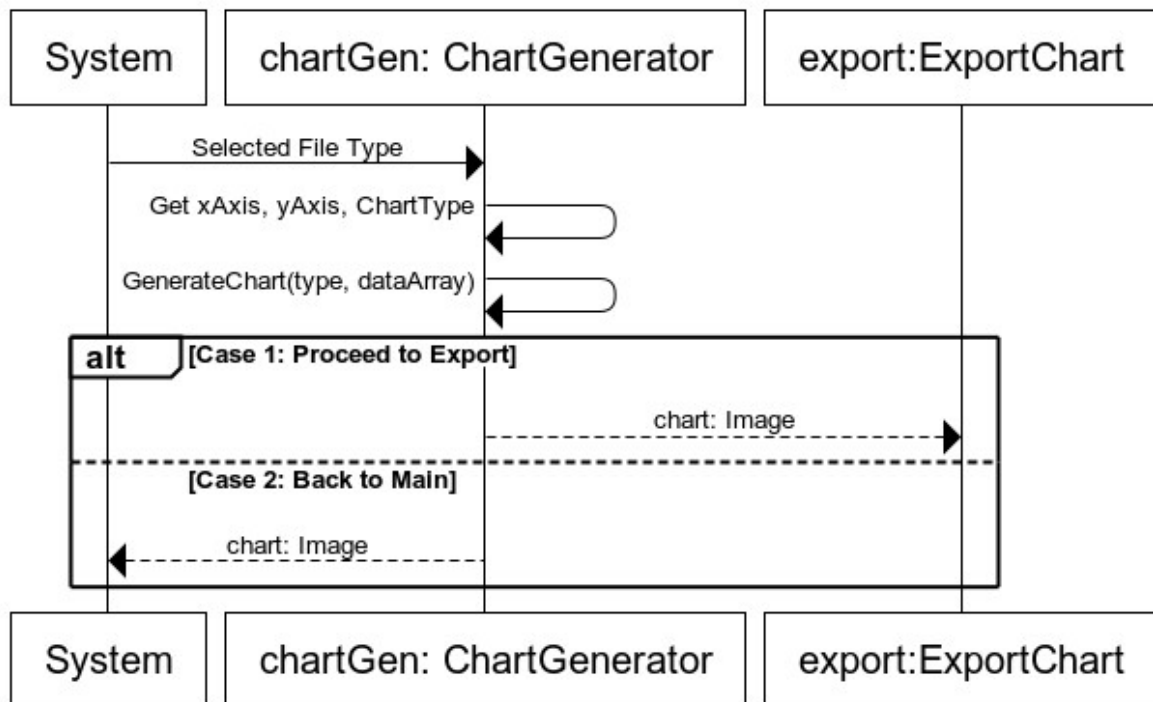
User Supplies File for Data



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Use Case 3: Generating a Chart

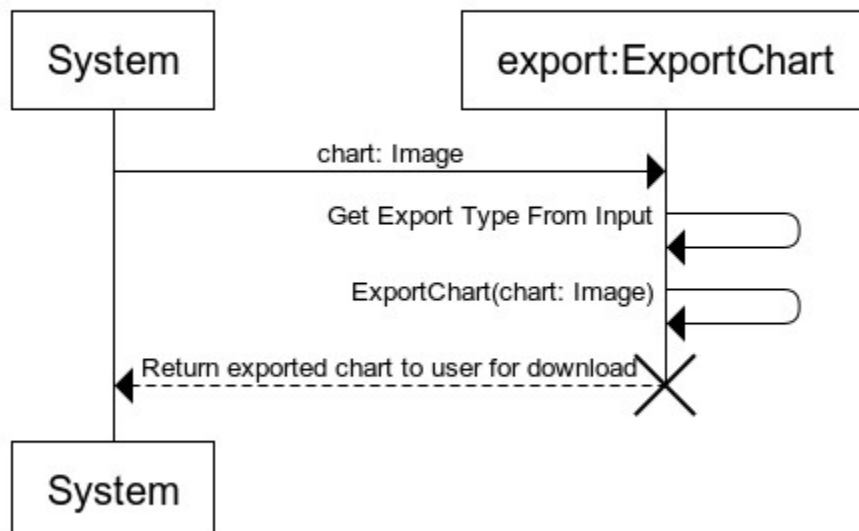
Chart Generation



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Use Case 4: Exporting a Generated Chart

Exporting a Chart



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In a group of 3-5, write a series of Use Cases, a UML Use Case Diagram, a UML Class Diagram, and UML Sequence Diagrams for the following client statement:

I want a system that will generate different types of charts from supplied data. The data may be in any number of formats including JSON and YAML. From the data, it will generate a line chart or scatterplot, possibly others. Charts should be able to be output into different image formats such as SVG or PDF.”

- Use Cases
 - Form a series of Use Cases that will satisfy the client statement (hint: there should be at least 4)
 - The use case must be in the Fowler Style, however, you may extend the Fowler format if necessary
 - All parts of the Fowler style must be present (Title, Main Success Scenario, Extensions)
- Use Case Diagram
 - Write a UML Use Case Diagram for your Use Cases
 - Diagram can be made using a tool or hand drawn
- Class Diagram
 - Draw a UML Class Diagram that shows the design for a system that will satisfy the client statement
 - Diagram can be made using a tool or hand drawn
- Sequence Diagram
 - Draw UML Sequence Diagrams that illustrate the Use Cases (as many diagrams as there are Use Cases, i.e., at least 4) and will satisfy the client statement
 - Diagrams can be made using a tool or hand drawn